

Time evolution operator

We seek a time-evolution operator $U(t, t_0)$ such that

$$\psi(x, t) = U(t, t_0)\psi(x, t_0)$$

By the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = H(t)\psi(x, t)$$

Substitute for $\psi(x, t)$.

$$i\hbar \frac{\partial}{\partial t} [U(t, t_0)\psi(x, t_0)] = H(t)U(t, t_0)\psi(x, t_0) \quad (1)$$

Noting that

$$\frac{\partial}{\partial t} \psi(x, t_0) = 0$$

we have

$$i\hbar \frac{\partial}{\partial t} [U(t, t_0)\psi(x, t_0)] = i\hbar \left[\frac{\partial}{\partial t} U(t, t_0) \right] \psi(x, t_0) \quad (2)$$

By the equivalence of (1) and (2)

$$i\hbar \left[\frac{\partial}{\partial t} U(t, t_0) \right] \psi(x, t_0) = H(t)U(t, t_0)\psi(x, t_0)$$

By operator equivalence

$$i\hbar \frac{\partial}{\partial t} U(t, t_0) = H(t)U(t, t_0)$$

Rewrite as

$$\frac{\partial}{\partial t} U(t, t_0) = -\frac{i}{\hbar} H(t)U(t, t_0)$$

Rearrange as

$$\frac{1}{U(t, t_0)} \partial U(t, t_0) = -\frac{i}{\hbar} H(t) \partial t$$

Integrate both sides.

$$\int_{t_0}^t \frac{1}{U(\tau, t_0)} dU(\tau, t_0) = -\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau$$

Solve the first integral.

$$\log U(t, t_0) - \log U(t_0, t_0) = -\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau$$

By $U(t_0, t_0) = 1$ the second log drops out.

$$\log U(t, t_0) = -\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau$$

Hence

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau \right) \quad (3)$$

Eigenmath script